

Tide retrospective: *Sextic 1D moments*

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Abstract

A single Tide-loop excursion in the `laplace` seabed, generalising the pure-quartic 1D file from $L(x) = x^4/24$ to $L(x) = x^6/720$. The headline theorem (`sextic_cov_affine`) gives the affine covariance of the 1D pure-sextic Gibbs measure as $\text{Cov}_t[ax + c, bx + d] = ab \cdot (720/t)^{1/3} \cdot \Gamma(1/2)/\Gamma(1/6)$, a rate- $t^{-1/3}$ analogue of the quartic's $t^{-1/2}$. The new file (396 lines) parallels its quartic predecessor section by section. Zero `sorry`, zero `axiom`, zero `native_decide`; clean Lean diagnostics on first pass. The interesting features of this run were two tactical refinements suggested by GPT-5.5 Pro at the deliberation step — a *B+ refinement* stating the half-line moment lemma for arbitrary natural power m rather than just $2n$, and a piecewise (rather than discriminant-based) proof of the polynomial bound $x^2 \leq 1 + x^6$ used for Gaussian-comparison integrability — both of which made the file genuinely cleaner than the quartic template it descended from. Like the concurrent 2D pure-quartic excursion, this run hit a working-tree race when another agent rewrote the top-level import file in the staging window; the fix was atomic re-staging plus a branch-ref re-pointing dance.

1 Setting

This Tide step is the sixth excursion outward from the same seabed: the `laplace` repo, which originally formalised the 1D anharmonic asymptotic $\text{Cov}_t[x^2, x] = -2\alpha/(\lambda^3 t^2) + o(t^{-2})$ (Susceptibility Primer, eq. 4.10) and has since been extended through five prior tide steps.¹ The first four merged into `main` on 1 May (commit 44a6001); this step shares its base with the parallel 2D pure-quartic run and was branched off the latter's tip per the Tide skill's branch-hygiene rule.

The candidate was picked from a cross-tide candidate survey written earlier the same day.² Of the five candidates listed there, *Sextic 1D moments* scored as the third choice, behind the threepoint κ_3 candidate (different seabed) and the 2D pure-quartic case (the parallel run). It is the cleanest test of whether the substitution-and- Γ template established by the quartic file generalises beyond a single exponent class. The pattern $L = x^{2k}/(2k)!$ predicts a rate $t^{-1/(2k-1)}$, with $k = 2$ giving the quartic's $t^{-1/2}$ and $k = 3$ giving the sextic's $t^{-1/3}$; this tide is the second data point.

The Step 2 deliberation (Claude $\leftrightarrow$ GPT-5.5 Pro) drafted three candidates: A, the full sextic package paralleling the quartic file; B, just the atomic half-line moment as a kernel; C, the general $p = 2k$ family covering quartic and sextic uniformly. Both parties backed A, with GPT making two tactical refinements that were carried into Step 3. The full deliberation is in the tide log.³

¹Branches `tide/quartic-moments`, `tide/2d-semi-degenerate`, `tide/universal-scaling`, `tide/f1-large-t`, and the concurrent `tide/2d-pure-quartic`.

²`projects/automation/log/2026-05-02-tide-candidates-survey.md`.

³`projects/primer/tide-log/2026-05-02-tide-sextic-moments.md`.

2 The thing formalised

For the centred sextic 1D potential

$$L(x) = \frac{x^6}{720}$$

and affine observables $\varphi(x) = ax + c$, $\psi(x) = bx + d$, against the 1D Gibbs measure $e^{-tL(x)} dx / Z(t)$:

For all $t > 0$,

$$\text{Cov}_t[\varphi, \psi] = ab \cdot \left(\frac{720}{t}\right)^{1/3} \cdot \frac{\Gamma(1/2)}{\Gamma(1/6)} = ab \cdot \frac{\sqrt{\pi} (720/t)^{1/3}}{\Gamma(1/6)}.$$

The Lean signature:

```
theorem sextic_cov_affine {t : ℝ} (ht : 0 < t) (a b c d : ℝ) :
  gibbsCov sexticPotential t (fun x => a * x + c) (fun x => b * x + d) =
    a * b * (720 / t) ^ ((1 : ℝ) / 3) * Real.Gamma ((1 : ℝ) / 2) /
    Real.Gamma ((1 : ℝ) / 6)
```

located at `Laplace/OneD/Sextic.lean`, in namespace `Laplace.OneD`. The file commits twelve further lemmas in addition to the headline: the potential definition, two flavours of integrability, the atomic half-line moment for arbitrary natural power, even and odd full-line moments, the partition function and its positivity, even and odd expected values, and the $n = 1$ specialisation $\langle x^2 \rangle_t = (720/t)^{1/3} \Gamma(1/2) / \Gamma(1/6)$ which is the load-bearing input to `sextic_cov_affine`.

The shape $L(x) = x^{2k} / (2k)!$ means that the unique critical point $x = 0$ has $L^{(j)}(0) = 0$ for $j = 0, \dots, 2k - 1$ and $L^{(2k)}(0) = 1$: this is the rank-zero degenerate case, analogous to the quartic ($k = 2$) but more degenerate. The Hessian-based Laplace machinery from the rest of the seabed does not apply; instead the substitution $u = (t/720)^{1/6} x$ reduces every moment integral exactly to a Γ -function value, and the rate $t^{-1/3}$ falls out of the $1/(2k - 1)$ formula at $k = 3$.

Numerical sanity check before formalising: `scipy.integrate.quad` agreed with the closed forms to $\sim 10^{-12}$ relative error across $t \in \{0.1, 1.0, 10.0, 100.0\}$ and $n \in \{0, 1, 2, 3\}$, for the half-line moments, the full-line even moments, the partition function, the expected values, and the affine covariance. The bonus integration-by-parts identity $\langle x^6 \rangle_t = 120/t$ (independent of all the special-function constants) provides a Gibbs-side cross-check that is also clean.

3 Proof strategy

The atomic kernel of the file is the half-line moment `integral_pow_mul_exp_neg_sextic_Ioi`, stated for arbitrary natural power m :

$$\int_0^\infty x^m e^{-tx^6/720} dx = \frac{1}{6} \left(\frac{720}{t}\right)^{(m+1)/6} \Gamma\left(\frac{m+1}{6}\right).$$

This is a direct application of Mathlib's master gamma-integral lemma⁴ with $p = 6$, $q = m$, $b = t/720$, plus a conversion between $\mathbb{N}$ - and $\mathbb{R}$ -powers via `rpow_natCast` and a single $(t/720)^{-(m+1)/6} \rightarrow (720/t)^{(m+1)/6}$ inversion via `inv_rpow`. Stating this for all m (the *B+ refinement*) costs nothing extra in the proof and makes the even and odd full-line cases share a single substitution argument.

⁴`integral_rpow_mul_exp_neg_mul_rpow`.

From the half-line lemma the rest of the file is mechanical. The full-line even moment doubles via `integral_comp_abs` after rewriting the integrand in $|x|$ -form (this uses $x^{2n} = |x|^{2n}$ and $x^6 = |x|^6$, both via `sq_abs`); the odd moment vanishes via the same parity argument the quartic file used, `Odd.neg_pow` $\langle n, \text{rfl} \rangle$ together with $(-x)^6 = x^6$ to flip the sign of the integrand. The partition function is the $n = 0$ specialisation of the even moment, the expected value is the ratio of the moment by the partition function (with the cancellation $(720/t)^{(2n+1)/6-1/6} = (720/t)^{n/3}$), and the affine covariance expands $(ax + c)(bx + d)$, uses linearity of `gibbsExpectation` (which itself is unfolded against `integral_add` and `integral_const_mul` after building integrability witnesses for $1, x, x^2$ against the Gibbs weight), and collapses via parity of the $\langle x \rangle$ terms to $ab\langle x^2 \rangle_t$.

The integrability of $x^n e^{-tx^6/720}$ against Lebesgue measure is the only place the proof has any choices. The quartic file used an exact completing-the-square identity $t \cdot x^4/24 - x^2 = (tx^2 - 12)^2/(24t) - 6/t$ to dominate by a Gaussian with $b = 1$ and prefactor $e^{6/t}$. The sextic analogue would need to minimise a cubic $t \cdot x^6/720 - x^2$, which is messy in Lean. The route taken here, on GPT-5.5 Pro’s tactical advice, is the slick polynomial bound

$$x^2 \leq 1 + x^6$$

proved piecewise: on $|x| \leq 1$, $x^2 \leq 1 \leq 1 + x^6$; on $|x| \geq 1$, $x^4 \geq 1$ so $x^2 \leq x^2 \cdot x^4 = x^6 \leq 1 + x^6$. Multiplying through by $t/720 > 0$ gives $t \cdot x^6/720 \geq (t/720)x^2 - t/720$, hence $e^{-tx^6/720} \leq e^{t/720} \cdot e^{-(t/720)x^2}$, dominated by a Gaussian of width $t/720$. Mathlib’s `integrable_rpow_mul_exp_neg_mul_sq` handles the dominator directly with $b = t/720$; the rest is bookkeeping. The piecewise lemma is nine lines; its quartic counterpart was thirteen.

4 Roadblocks and resolutions

None of mathematical substance. The proof typechecked clean on the first compile after a one-line tactic fix (`rcases le_or_lt` $\rightarrow$ `rcases le_total`; `le_or_lt` no longer exists in the version of Mathlib pinned by the seabed) and a one-line cleanup of an unused `push_cast` that had been a vestige of copy-pasting from the quartic file. The candidate had been picked precisely so that the seabed’s machinery would already cover the moves; the run paid the cost of that prediction.

Concurrent-agent staging race. Same root cause as the parallel 2D pure-quartic Tide. Multiple times during this run, the working-tree copy of `Laplace.lean` was overwritten by the other agent who was also adding an import for their new file. The interesting variant, in this run, was that the `git commit` fired while the working tree was on the *wrong branch*: at the moment of commit, the parallel agent had checked out their `tide/2d-pure-quartic` branch in the shared working tree, so my Sextic commit landed on *their* branch rather than my own. The fix was

```
git branch -f tide/sextic-moments HEAD      # claim the commit
git checkout tide/sextic-moments
git branch -f tide/2d-pure-quartic HEAD~1   # restore their tip
```

which by happy accident produced exactly the linear-stack arrangement the Tide skill’s branch-hygiene rule requires (Sextic on top of PureQuartic on top of `main`), but it was a recovery move rather than a deliberate one. After the parallel run later added a trailing `Laplace.lean` import-list fix-up commit and a retrospective commit on top of their original tip, this branch was rebased onto the new tip via `git rebase tide/2d-pure-quartic`, which applied without conflict.

The structural lesson is the same as the parallel retrospective concluded: concurrent Tide agents need filesystem isolation. `git worktree` per Tide branch is the right answer.

Tactical advice from GPT-5.5 Pro that paid off. Two pieces of GPT-5.5 Pro’s deliberation-stage response were carried into the proof and each paid for itself.

- *The B+ refinement.* GPT noted that on Ioi0, parity is irrelevant, so stating the half-line moment for arbitrary natural power m costs nothing extra and makes the file’s even/odd full-line story share a single substitution argument. This is the kind of strict-generalisation-for-free that Tide-loop deliberation is for; carrying it in saved a near-duplicate half-line argument for the odd case.
- *Piecewise integrability.* GPT pre-empted the obvious sin (which I had drafted) of proving $x^2 \leq 1 + x^6$ via a cubic-discriminant argument:

GPT-5.5 Pro: Your proposed bound $x^2 \leq 1 + x^6$ is excellent, but in Lean I would **not** prove it via discriminant/minimum. I would prove it piecewise: if $|x| \leq 1$, then $x^2 \leq 1 \leq 1 + x^6$; if $1 \leq |x|$, then $x^2 \leq x^6$, hence $x^2 \leq 1 + x^6$. That will be much friendlier than a cubic-minimum argument.

The piecewise version that landed is nine lines, three of which are `nlinarith`-with-hint closers; the discriminant version would have been a forty-line minimisation by hand. This was the highest-value tactical advice in the deliberation.

Beyond these two, no proof-side roadblocks: the file is essentially a substitute-and-rename of `Quartic.lean` with the integrability lemma swapped out and the half-line lemma generalised in m .

5 What was Mathlib, what was new

Mathlib provided. `integral_rpow_mul_exp_neg_mul_rpow` for the master half-line gamma integral. `integrable_rpow_mul_exp_neg_mul_sq` for the Gaussian dominator. `integral_comp_abs` for the even-symmetric reduction to the half line. `integral_neg_eq_self` for the odd parity argument. The standard `integral_add` / `integral_const_mul` / `Integrable.const_mul` / `add` lemmas for the affine-observable expansion. `rpow_natCast`, `inv_rpow`, `rpow_neg`, `Real.rpow_add` for exponent algebra. `Real.Gamma_pos_of_pos` for partition-function positivity.

The seabed provided. `Laplace.Gibbs` (partitionFunction, gibbsExpectation, gibbsCov for the 1D Gibbs interface), and `Laplace/OneD/Quartic.lean` as the structural template. No new seabed-side infrastructure was needed.

What was new in this tide. The 1D pure-sextic potential, a Gaussian-comparison integrability lemma (with the piecewise polynomial bound $x^2 \leq 1 + x^6$ as a private helper), the half-line moment lemma in the B+ form for arbitrary natural power, the full-line even and odd moments, the partition function and its positivity, the four expected-value lemmas (even, odd, $n = 1$, linear specialisation), and the headline affine covariance.⁵ There is no novel mathematical content; what is new is the substitution-into-the-template that confirms the substitution template ports cleanly to a second exponent class. As the parallel 2D pure-quartic retrospective predicted in its *Lessons*: “the next test is whether [the factorisation template] ports across the sextic side; if so, the abstraction is genuinely the right shape.” It does, and it is.

⁵Lean names: `sexticPotential`; `sextic_integrable_pow`; `sq_le_one_add_pow_six` (private); `integral_pow_mul_exp_neg_sextic_Ioi`; `sextic_moment_even`, `sextic_moment_odd`; `sextic.partition`, `sextic.partition_pos`; `sextic.expected_value_even`, `_odd`, `_sq`, `_lin`; `sextic.cov_affine`.

6 Lessons

- *Specific tactical advice from GPT-5.5 Pro at the deliberation stage transfers cleanly into Step 3.* The B+ refinement and the piecewise integrability bound were both proposed at Step 2 and carried verbatim into the final proof. This is the right division of labour: the deliberation stage is where “how you should phrase the lemma” and “how you should prove the polynomial inequality” get decided, before any Lean is written. The cost was a single 5-minute consult; the value, fifty lines of avoided proof.
- *The substitution-and- Γ template is robust at $k = 3$.* The $L = x^{2k}/(2k)!$ family produces, at each k , a rate $t^{-1/(2k-1)}$ covariance with constants in $\Gamma((2n+1)/(2k))/\Gamma(1/(2k))$ and a proof that mirrors the previous k section by section. With $k = 2$ and $k = 3$ both done in essentially the same shape, candidate C from the deliberation (the unifying $p = 2k$ parameterisation) becomes substantially more attractive than it did at $k = 2$ alone — the case for “just refactor and parameterise” has gone from one data point to two.
- *Concurrent Tide agents in the same checkout fight over `Laplace.lean`.* Same lesson as the parallel retrospective draws. Filesystem isolation via `git worktree` is the right structural fix; ad-hoc atomic re-staging is at best a recovery move.
- *Piecewise polynomial bounds beat discriminant-based ones in Lean.* The `nlinarith`-with-hint closers do not like cubic minima, but a case split on $|x| \leq 1$ vs $|x| \geq 1$ followed by a single `nlinarith` per branch is robust. Worth lifting to the seabed’s `CLAUDE.md` as a tactic note.

7 Follow-ups

- *The $p = 2k$ parameterisation, now plausible.* With `Quartic.lean` ($k = 2$) and `Sextic.lean` ($k = 3$) now both in place and structurally near-identical, the unifying lemma

$$\int_0^\infty x^m e^{-tx^{2k}/(2k)!} dx = \frac{1}{2k} \left(\frac{(2k)!}{t} \right)^{(m+1)/(2k)} \Gamma\left(\frac{m+1}{2k}\right)$$

is now reachable in roughly the same proof body as the present file, just with k generic. Best opened as a future Tide step: not a refactor of the existing files (they stay as worked examples), but a new `Laplace/OneD/PolyEven.lean` (or similar) that proves the parameterised identity once and recovers the quartic and sextic specialisations as corollaries. Candidate B and C from the present deliberation would land together as part of that step.

- *Quartic–sextic mixed potential.* $L(x, y) = x^4/24 + y^6/720$, with rates $t^{-1/2}$ in x and $t^{-1/3}$ in y . Mechanical given the present file plus `Laplace/TwoD/PureQuartic.lean` from the parallel Tide, particularly if the additively-separable factorisation refactor (deferred in that retrospective) lands first.
- *Mathlib upstreaming for polynomial-times-Gibbs integrability.* `quartic_integrable_pow` and `sextic_integrable_pow` are the same shape: $\int x^n \exp(-b \cdot x^{2k})$ for $b > 0, k \geq 2, n \in \mathbb{N}$. With two concrete uses, the case for promoting a generic version to `lean-common` (or upstream into Mathlib alongside the existing `integrable_rpow_mul_exp_neg_mul_sq`) is now stronger than at one use.
- *\leanref tagging.* If the primer gains a sextic-rate paragraph in its degenerate-case discussion, tag it with `\leanref{sextic_cov_affine}`. Currently no such paragraph exists; the

sextic case is implicit in the $L = x^{2k}/(2k)!$ generalisation that the primer alludes to but does not spell out. Worth a one-paragraph appendix addition the next time the primer’s degenerate-case section is touched.

- *Worktree-isolated Tide runs.* Same as the `tide/2d-pure-quartic` retrospective notes: a small wrapper that creates a `git worktree` per Tide branch and runs the agent against that worktree would prevent the staging race. Worth adding to the Tide skill as a preflight setup step.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the B+ refinement of the half-line moment lemma and the piecewise proof of the polynomial bound $x^2 \leq 1 + x^6$. Both were carried into the final proof, and both were the right call. The full consult and the tide log (with candidate drafts, vote, numerical check, and result) are both preserved in the primer project’s tide-log directory.⁶ The parallel 2D pure-quartic run shared the seabed and the integration race; its retrospective is the immediate predecessor to this one and recorded the same lessons from the other side.

⁶`projects/primer/tide-log/gpt55_tide_sextic_v1.md`
`2026-05-02-tide-sextic-moments.md`.

and `projects/primer/tide-log/`